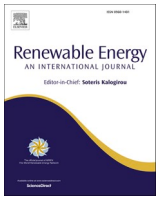
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Innovative passive cooling of photovoltaic panel using loop heat pipe technology with passive daytime radiative cooling

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ABSTRACT

This paper proposes an innovative cooling technique that utilizes a loop heat pipe (LHP) and passive daytime radiative cooling. The proposed system uses LHP to move the heat load from the photovoltaic (PV) panel surface facing the Sun to a radiator surface where heat is radiated to the sky and dissipated to the surroundings by convection. The LHP allows heat removal from PV panel surface to the radiator surface with negligible temperature difference. Based on selected PV parameters and conditions, the proposed LHP-radiator cooling system achieves a peak temperature reduction of over 10 °C, leading to a 4.6 % enhancement in PV electrical efficiency compared to the reference efficiency. The selected systems' results show an annual energy saving of 3.4 % (around 8.7 kWh/m^2) for the year of 2022.

1. Introduction

The dramatically growing demand for green energy is driven by the need for energy in daily activities and the necessity to reduce the harmful impact of fossil fuels on global climate change. The recent severe climate events and their fluctuation have altered people's demands. The record global temperatures, severe rainfalls, extreme droughts, high wind speed, and increase in wildfires are strongly related to the use of fossil fuels. Recently, more people started demanding green energy to mitigate these extreme climate swings. In the UAE, the demand for green energy has led to many new renewable energy projects such as the Mohammed bin Rashid Al Maktoum Solar Park in Dubai and Shams-1 project in Abu Dhabi. The use of photovoltaic (PV) panels keeps growing due to the strong demand and government regulations. Investments in PV solar farms were $114 billion during 2018 and it is expected to reach $165 billion by 2030 and $192 billion by 2050 [1]. The demand for green energy has fostered many research developments and innovations in the field of solar energy [2–8].

One of the successful solar energy technologies is the use of PV panels to convert solar irradiance (mainly visible light) to electrical energy. However, PV panels are prone to overheating which is a critical problem that impacts the performance of regular and concentrated PV panels. Overheating in PV panels, which hinders the performance of PV, results from high ambient temperatures, excessive irradiance and inappropriate cooling of the PV panels [9]. Cooling is an important design item that needs to be considered when building solar PV farms, especially in the Gulf countries which receive high solar irradiation and fall under scorching summer temperatures. The use of a loop heat pipe (LHP) in cooling PV is limited and more research is needed in this area [10]. On the other hand, the use of forced cooling has been examined by many research groups [11–15].

Many ideas have been proposed to keep the PV panels' temperatures under control such as using natural air cooling [16,17], liquid water cooling [9], clay pot evaporative cooling [18], phase-change material (PCM) cooling [19], heat pipe [20] and loop heat pipe [21]. It has been shown in literature [16] that convection cooling is able to shave off a 5 °C when tested in Turkey. Passive natural air convection cooling has been used to cool concentrating PV panels and has shown a 30 °C drop when micro fins were used. The temperature dropped from 100 to 70 °C by offering a bigger area under natural convection conditions [9]. Spraying water on PV panels has enhanced PV cooling which has led to improvement in PV performance [9]. Based on a study conducted in Egypt, spraying water when the PV surface temperature reaches 45 °C offers an enhancement in the PV panels' performance, while consuming a minimum amount of water. Using evaporative cooling has shown an

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Nomenclature

A	area [m^2]
B	temperature coefficient
C	fraction of the sky covered by clouds
$\dot{D}$	diffuse radiation [W/m^2]
$\dot{E}$	electrical power [W]
F	view factor
$\dot{G}$	total solar irradiance [W/m^2]
Gr	Grashof number
h	convection coefficient [$W/(m^2 \bullet^\circ C)$]
HP	Heat pipe
$\dot{I}$	direct normal irradiance [W/m^2]
K	scale factor based on cloud height
L	characteristic length
LHP	Loop heat pipe
Nu	Nusselt number
Pr	Prandtl number
PCM	Phase-change material
$\dot{q}_{sky}$	thermal downwelling irradiation [W/m^2]
$\dot{Q}$	heat transfer [W]
RH	relative humidity
Ra	Rayleigh number
RC	Radiator cooling
T	temperature [$^\circ C$]
g	gravity acceleration [m/s]
k	air thermal conductivity [$W/(m \bullet^\circ C)$]

Greek letters

α	absorptivity
β	air thermal expansion
ε	emissivity
η	efficiency
γ	tilt angle
θ	angle between the plate and normal axis
ν	air kinematic viscosity
ρ	Reflectivity coefficient
σ	Stefan–Boltzmann constant

Subscripts

a	air
amb	ambient
b	bottom
c	convection
grd	ground
L	longwave irradiance
LHP	Loop heat pipe
PV	photovoltaic
r	radiation
R	radiator
ref	reference condition
s	surface
sky	sky
S	shortwave irradiance
t	top

8 °C drop in the surface temperature of PV panels when tested under the conditions of Chidambaram City in India [18]. While testing the PV-PCM system in Slovenia [19], results have shown around a 14 °C drop in PV temperature when compared to an uncooled PV unit (i.e., cooled by natural air). Also, the study showed that for a period of one year, PCM cooling could improve PV performance by 7.3 %. As demonstrated in the literature [22], the use of PCM has improved the performance of PV-thermal systems. Modjinou et al. [23] investigated the improvement in the photovoltaic/thermal (PV/T) system when it was fitted with macro-encapsulated PCM and micro-channel heat pipes. The study [23] showed that such improvement has raised the combined efficiency of a regular PV/T system and has prevented freezing incidents in the solar water collector. A research team in Spain [24] has examined a patent design which utilizes a PV-HP-PCM system, where a heat pipe (HP) is used to enhance heat transfer between PV and PCM. Zhang et al. [21] have experimentally tested a photovoltaic/loop-heat-pipe heat pump system and showed an improvement of 3–5% in solar exergetic efficiency when compared to standard PV systems.

Heat pipe (HP) and Loop heat pipe (LHP) technologies have been used in electronics cooling for years. Both devices can transfer a large amount of heat with a minute temperature difference. Historically, LHP was developed after HP to achieve long-distance heat transfer. HP offers a robust way to remove a large amount of heat from electronics and prevent electronics from overheating [25–27]. LHP is an attractive technique that has been developed for space applications [28–30], which later gained more ground in electronics cooling. In a LHP, liquid and vapor move in a loop in the same direction, while in HP, vapor and liquid move in the opposite direction. The working liquid absorbs a large amount of thermal energy and changes phase to vapor. Then the produced vapor is pushed toward the condenser, where it gets condensed. The condensed liquid is seeped back to the hot source by the capillary force through a wick porous substrate.

This study is examining the benefit of using a loop heat pipe with radiator cooling (LHP-RC) to increase the amount of heat transfer by extending the radiation surfaces. The LHP allows moving heat from the PV to the radiator with negligible temperature difference (due to its small thermal resistance). Having the radiator in a shadow (or a low solar irradiance zone) allows radiating the heat to the sky, which is cooler than the ambient temperature. The study explores the feasibility of the proposed PV-LHP-RC system to operate in Sharjah city in the United Arab Emirates (UAE). In addition, the study utilizes the developed mathematical model in predicting the system performance under different design parameters such as radiator area, backside convection coefficient, radiator emissivity and LHP thermal resistance.

2. Problem formulation and mathematical analysis

A schematic diagram of the proposed PV-LHP-RC system is shown in Fig. 1 with heat transfer modes into and out of the system. The PV panel tilt angle (γ) is selected as 25°, which is the recommended value for PV installation in the University City in Sharjah, UAE. Since the analysis is performed for Sharjah, the PV system is oriented due South so it is exposed to solar irradiation through the day, while the radiator faces due North so it is not exposed to solar irradiation all day. A LHP is installed to transfer heat from the PV panel to the radiator side of the system to improve heat removal and lower the PV panel temperature. To assess its performance, an analysis has been performed for the proposed system and compared to a single PV panel without a cooling system. The heat transfer from a single PV, shown in Fig. 2, under natural convection and radiation is modeled and analyzed. As shown in Fig. 2, a single PV is fixed on a metallic frame with similar orientation used in the proposed PV-LHP-RC system.

The total solar irradiance ($\dot{G}_H$) refers to the total power per unit area of solar irradiation received on a horizontal surface on the ground. It consists of both direct normal irradiance ($\dot{I}$) and diffuse radiation ($\dot{D}$). The diffuse radiation refers to solar radiation that reaches the Earth's surface after scattering in the atmosphere. It occurs when sunlight interacts with air molecules, water droplets, and aerosols, causing the

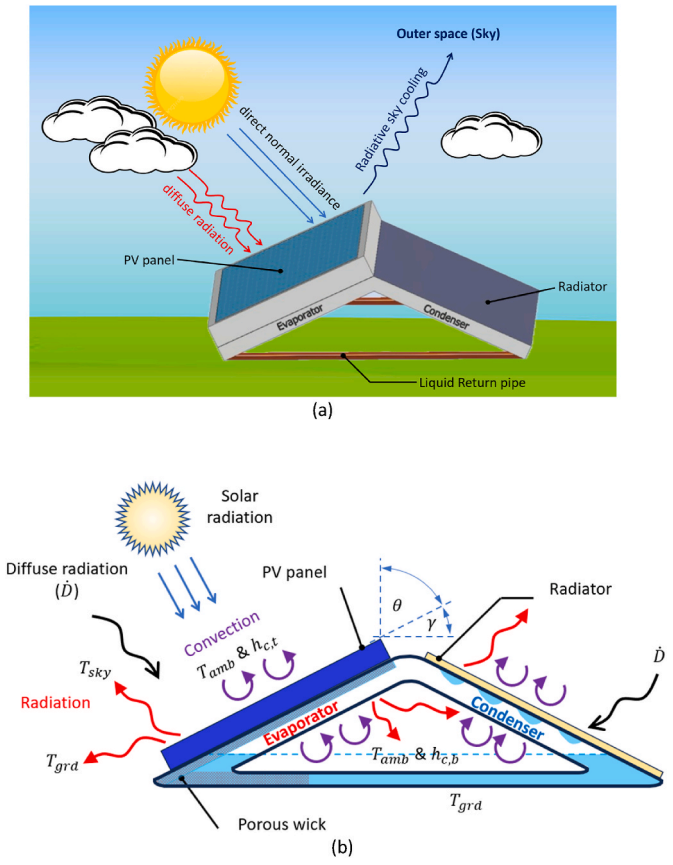


Fig. 1. The proposed system illustrates in (a) a 3D CAD drawing showing radiative heat transfer modes, including outer-space radiation, direct normal radiation and diffusive radiations, and (b) a 2D schematic side-view diagram of the proposed PV-LHP-RC system.

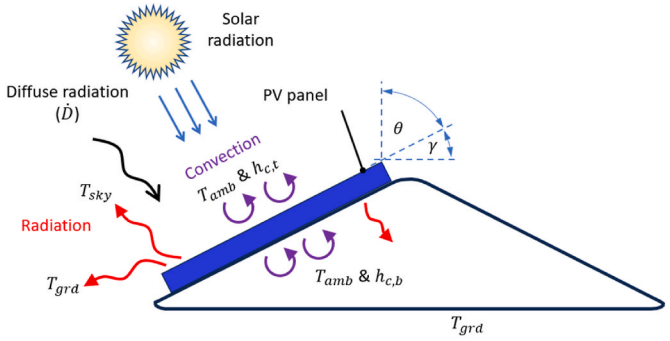


Fig. 2. A schematic side-view diagram of a single photovoltaic system under natural convection and radiation.

sunlight to alter its direction. The solar irradiance can be expressed as:

$$\dot{G}_H = \dot{I}_H + \dot{D} \tag{1}$$

In this study, the solar irradiance on a horizontal surface ($\dot{G}_H$) has been obtained from the NASA Langley Atmospheric Science Data Center (POWER Projects) [31]. NASA POWER (Prediction of Worldwide Energy Resource) projects provide data that is based on a combination of measurements and models. Two sets of data that are important to comprehend are "all-sky surface irradiance" and "clear-sky surface irradiance". All-sky surface irradiance refers to the total solar irradiance reaching the Earth's surface under any sky condition, including both direct sunlight and diffuse radiation, regardless of cloud coverage. It accounts for the effects of clouds, atmospheric particles, and other factors that may scatter or absorb sunlight. This value is used in the mathematical calculations, and it is designated with the symbol $\dot{G}_H$. Clear-sky surface irradiance is the solar irradiance that would be received at the Earth's surface in the absence of clouds and other atmospheric obstructions. It represents the theoretical maximum solar radiation under clear atmospheric conditions which is not used in this analysis.

On clear days with minimal cloud cover, the ratio of diffuse to total solar radiation might be around 10–20 % or even less, depending on atmospheric conditions. Direct sunlight from the sun's direct path is a significant portion of the total solar radiation. While on cloudy or overcast days, the ratio of diffuse to total solar radiation can increase significantly, and diffuse radiation may constitute most of the total. In some cases, the ratio may exceed 50 %, and even higher values are possible under heavy cloud cover. Since this value is not available in the POWER projects data and Sharjah is known for its clear sky most of the year, in this study, the diffuse radiation to the total is set to 10 %.

2.1. Radiation on tilted surfaces

Given that the proposed photovoltaic modules are set at a fixed tilt, there is a need to estimate the solar radiation on these tilted surfaces (PV panel and radiator) based on horizontal measurements. This estimation involves considering separate components for direct and diffuse irradiation. A simpler model is favored; hence the Telecom method [32] is adopted in this study. In cases where insolation data is accessible in the format of direct and diffuse components, the subsequent method, as outlined by Mack in 1979 [33] and presented in Refs. [32,34], can be employed to ascertain the corresponding insolation received by a solar panel tilt angle (γ) relative to the horizontal. It is appropriate to assume the diffuse irradiance ($\dot{D}$) is independent of the tilt angle. The direct irradiance on the horizontal surface ($\dot{I}_H$) needs to be transformed into the direct component incident on a plane tilted at an angle γ to the horizontal, as depicted in Fig. 3.

Using Fig. 3, the directional irradiance incident on the PV panel ($\dot{I}_{PV}$) is given as shown below [32–34]

$$\dot{I}_{PV} = \frac{\dot{I}_H \sin(\phi + \gamma)}{\sin \phi} \tag{2}$$

Using Fig. 3, one can show that the directional irradiance incident on the radiator ($\dot{I}_R$) can be calculated as follow:

$$\dot{I}_R = \frac{\dot{I}_H \sin(\phi - \gamma)}{\sin \phi} \tag{3}$$

Where γ and ϕ are the tilt angle and the altitude angle. The altitude angle (Sun angle) represents the angle between the sun and the horizontal and is given as [32,34]:

$$\phi = 90 - \varphi + \delta \tag{4}$$

where φ is the northern latitude and δ is the declination angle. The northern latitude for Sharjah is $\varphi = 25.348766$. The declination angle is given as follow [34]:

$$\delta = 23.45^{\circ} \sin\left[\frac{360}{365}(284 + d)\right] \tag{5}$$

where d is the day of the year in Julian dates format.

Hence the total solar irradiance falling in the PV and radiator are written as shown below:

$$\dot{G}_{PV} = \dot{I}_{PV} + \dot{D} \tag{6}$$

$$\dot{G}_R = \dot{I}_R + \dot{D} \tag{7}$$

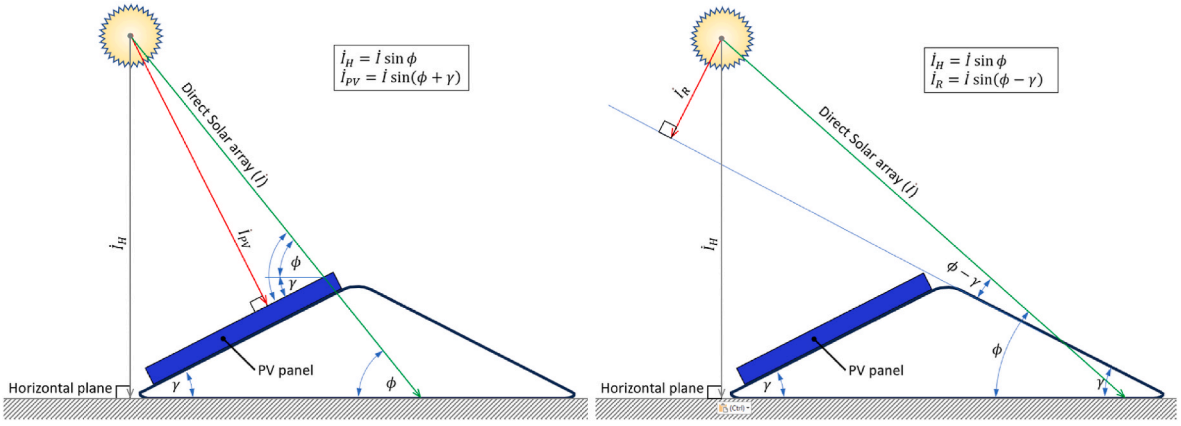


Fig. 3. Light incident on a surface tilted to the horizontal considering (a) PV panel and (b) Radiator surface.

Table 1
View factors used for different surfaces in the radiation calculation of the proposed system.

View factor	Definition	Calculations	Value
$F_{PV,sky}$	from PV top-surface to the sky	Depends on the view angle ratio = 155/180 = 0.862	0.861
$F_{R,sky}$	from radiator top-surface to the sky	Depends on the view angle ratio = 155/180 = 0.862	0.861
$F_{PV,grd}$	from PV top-surface to the ground	Depends on the view angle ratio = 25/180 = 0.139	0.139
$F_{R,grd}$	from radiator top-surface to the ground	Depends on the view angle ratio = 25/180 = 0.139	0.139
$F_{R,PV}$	from radiator back-surface to the PV panel backside	Three-sided enclosure	0.094
$F_{R,b,grd}$	from radiator back-surface to the LHP ground section	$F_{i\to j} = \frac{w_i + w_j - w_k}{2w_i}$	0.906
$F_{PV,R}$	from PV back-surface to the Radiator backside		0.094
$F_{PV,b,grd}$	from PV back-surface to the LHP ground section		0.906

2.2. Sky temperature

Sky temperature (T_{sky}) typically refers to the temperature of the Earth's atmosphere as observed or measured at a specific location and time. It is an atmospheric parameter that provides information about the thermal state of the air and its value is needed to calculate the thermal radiation from a surface to the sky. To calculate the sky temperature, one needs to calculate or measure the thermal downwelling irradiation ($\dot{q}_{sky}$). Thermal downwelling irradiation, or "longwave downward irradiance," refers to the downward-directed flow of thermal (infrared) radiation from the atmosphere towards the Earth's surface. The thermal downwelling irradiation (longwave) combined with solar irradiation (shortwave) forms the total sky irradiance (TSI). The majority of TSI is composed of shortwave irradiance ($\dot{G}$). The shortwave downwelling irradiation occurs exclusively during daytime hours, while thermal downwelling irradiation ($\dot{q}_{sky}$) is present continuously, extending from day to night.

Table 2
Photovoltaic panel main parameters.

Parameter	Definition	Value	Units
L	PV characteristic length (which is equal to the radiator characteristic length)	1.0	m
A	PV area (which is equal to the radiator area)	1	m^2
η_{ref}	PV reference efficiency	13.9 [15]	%
B_{PV}	PV temperature coefficient	0.045 [15]	$1/^\circ C$
T_{ref}	PV reference temperature	25 [15]	$^\circ C$

Table 3
Optical properties used in the simulation [44].

	Shortwave irradiance (Solar Spectrum) 0.3–3 μm	Longwave irradiance (Infrared Spectrum) 3-25 μm
PV emissivity/ absorptivity ($\varepsilon_{PV} = \alpha_{PV}$)	$\varepsilon_{PV,S} = \alpha_{PV,S} = 0.9$	$\varepsilon_{PV,L} = \alpha_{PV,L} = 0.85$
R emissivity/absorptivity ($\varepsilon_R = \alpha_R$)	$\varepsilon_{R,S} = \alpha_{R,S} = 0.01$	$\varepsilon_{R,L} = \alpha_{R,L} = 1.00$
PV reflectivity ($\rho_{PV} = 1 - \varepsilon_{PV}$)	$\rho_{PV,S} = 0.1$	$\rho_{PV,L} = 0.15$
R reflectivity ($\rho_R = 1 - \varepsilon_R$)	$\rho_{R,S} = 0.99$	$\rho_{R,L} = 0.00$

In 1963, Swinbank [35] presented a relation to calculate the thermal downwelling irradiation as a function of ambient temperature and relative humidity. In 2002, Goforth et al. [36] has modified the Swinbank relation by accounting for cloud height and percent cloud cover. The modified Swinbank (MS) relation, shown below, is a function of ambient air temperature at ground level, relative humidity, cloud height and cloud coverage.

$$\dot{q}_{sky} = \left(1 + KC^2\right) 8.778 \times 10^{-13} T_{amb}^{5.852} RH^{0.07195} \tag{8}$$

where K is a scale factor based on cloud height (ranging from 0.34 for very low clouds to 0.06 for very high clouds), C is the fraction of the sky covered by clouds, T_{amb} (in Kelvin) is the ambient temperature at ground level and RH is the surface relative humidity, as a percentage (e.g., RH = 20 in the case of 20 % relative humidity). This relation shows uncertainty of 4 % at a nominal air temperature of 15 $^\circ C$ and a relative humidity of 60 % [36]. The sky temperature (T_{sky}) can be calculated from thermal downwelling irradiation $\left(\dot{q}_{sky}\right)$ using the following relations [37].

$$T_{sky} = \left(\frac{\dot{q}_{sky}}{\sigma}\right)^{1/4} \tag{9}$$

As noted in the literature [38], the thermal downwelling irradiation $\left(\dot{q}_{sky}\right)$ is related to ambient temperature as follow:

$$\dot{q}_{sky} = \sigma T_{sky}^4 = \varepsilon_{sky} \sigma T_{amb}^4 \tag{10}$$

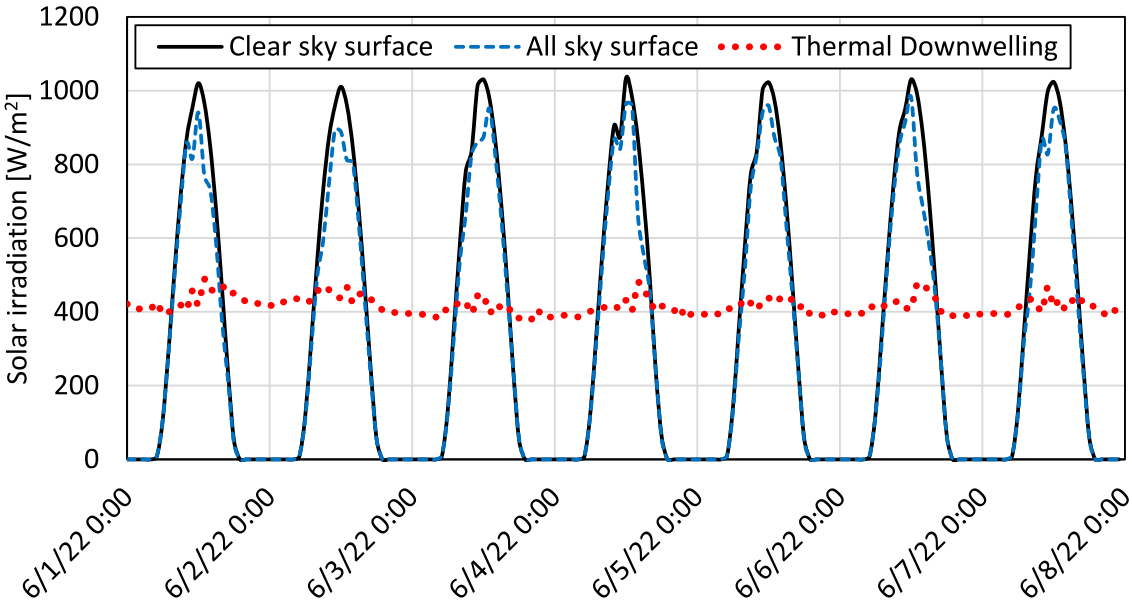


Fig. 4. Total solar irradiance for clear sky, total solar irradiance for all sky and thermal downwelling over a horizontal surface in the University City in Sharjah from 1-June-2022 till 8-June-2022.

$$T_{sky} = \varepsilon_{sky}^{1/4} T_{amb} \tag{11}$$

Where ε_{sky} is the emissivity of the atmosphere and σ is Stefan–Boltzmann constant. There are different models [38] to calculate the atmosphere emissivity (ε_{sky}). There is no need for these relations in this paper, since this work calculates the sky temperatures from thermal downwelling irradiation which is obtained from POWER projects and calculated from the MS relation.

2.3. LHP system

A loop heat pipe (LHP) is a two-phase flow heat transfer device that is capable of transporting a large amount of heat across a long distance with a small temperature difference. The temperature difference between the LHP evaporator and condenser sections depends on the LHP design and the heat load. A detailed thermodynamic analytical model of the LHP system used in this study can be found in Hamdan and Elnajjar [39], Hamdan et al. [28,30] and Hamdan [40]. A recent LHP study by Aono et al. [41] has reported a LHP thermal resistance of 0.004 °C/W at a heat load of 6200 W. In this study, the thermal resistance (R_{LHP}) is assumed constant with a value of 0.004 °C/W.

2.4. The proposed PV-LHP-RC system

As shown in Fig. 1, the energy transfer to the system occurs in two forms which are solar irradiance ($\dot{G}_{PV} = \dot{I}_{PV} + \dot{D}$) and thermal downwelling irradiation ($\dot{q}_{sky}$). At the PV, the absorbed solar irradiance by the PV is represented as $\alpha_{PV,S}(\dot{I}_{PV} + \dot{D})$, while the absorbed thermal downwelling irradiance is represented as $\alpha_{PV,L}\dot{q}_{sky}$, where $\alpha_{PV,S}$ and $\alpha_{PV,L}$ are PV absorptivity of shortwave and longwave irradiance, respectively. Part of the solar irradiance energy is absorbed by the PV and converted to electricity ($\dot{E}_{PV}$), while the rest of the energy is dissipated to the surroundings by convection ($\dot{Q}_{c,PV}$), radiation ($\dot{Q}_{r,PV}$) and absorbed by LHP in the evaporator section ($\dot{Q}_{LHP}$). The heat absorbed by the LHP at the evaporator ($\dot{Q}_{LHP}$) is transferred via the LHP to the LHP condenser and is released to the surrounding by convection ($\dot{Q}_{c,R}$) and radiation ($\dot{Q}_{r,R}$) at the radiator side of the system. At the radiator and under the assumption that the radiator is not exposed to solar irradiation all day, the solar irradiance absorbed by the radiator is represented as $\alpha_{R,S}\dot{D}$, while the thermal downwelling irradiance is represented as $\alpha_{R,L}\dot{q}_{sky}$, where $\alpha_{R,S}$ and $\alpha_{R,L}$ are radiator absorptivity of shortwave and longwave irradiance, respectively. Mathematically the energy balance of the entire system is written as shown below:

$$\left[\alpha_{PV,S}(\dot{I}_{PV}+\dot{D})+\alpha_{PV,L}\dot{q}_{sky}\right]+\left[\alpha_{R,S}(\dot{I}_{R}+\dot{D})+\alpha_{R,L}\dot{q}_{sky}\right]=\dot{E}_{PV}+\dot{Q}_{c,PV}+\dot{Q}_{r,PV}+\dot{Q}_{c,R}+\dot{Q}_{r,R} \tag{12}$$

The energy equation for amount of heat transfer by LHP is written by enforcing energy balance on the PV-LHP side or RC-LHP side, respectively, as shown below.

$$\left[\alpha_{PV,S}(\dot{I}_{PV}+\dot{D})+\alpha_{PV,L}\dot{q}_{sky}\right]=\dot{Q}_{LHP}+\dot{E}_{PV}+\dot{Q}_{c,PV}+\dot{Q}_{r,PV} \tag{13}$$

$$\left[\alpha_{R,S}(\dot{I}_{R}+\dot{D})+\alpha_{R,L}\dot{q}_{sky}\right]+\dot{Q}_{LHP}=\dot{Q}_{c,R}+\dot{Q}_{r,R} \tag{14}$$

Adding the last two equations produces the energy balance equation for the whole system (Eq. (12)).

The electrical power produced by the PV can be calculated as follows:

$$\dot{E}_{PV} = \eta \bullet \dot{G}_{PV} \tag{15}$$

where η is the PV electrical efficiency which depends on the PV information provided by the manufacturer, mainly the reference efficiency (η_{ref}), reference temperature (T_{ref}), and temperature coefficient (B_{PV}) [42].

$$\eta = \eta_{ref}\left[1 - B_{PV}\left(T_{PV} - T_{ref}\right)\right] \tag{16}$$

The convection and radiation heat transfer at the PV surface can be calculated using the following equations, where heat is lost from top- and bottom-surfaces. The radiation heat transfer is radiating to the surroundings which comprise from the sky and ground. To simplify the analysis, all the surfaces of the PV panel are assumed at uniform temperature of T_{PV}.

$$\dot{Q}_{c,PV} = h_t A_{PV}(T_{PV} - T_{amb}) + h_b A_{PV}(T_{PV} - T_{amb}) \tag{17}$$

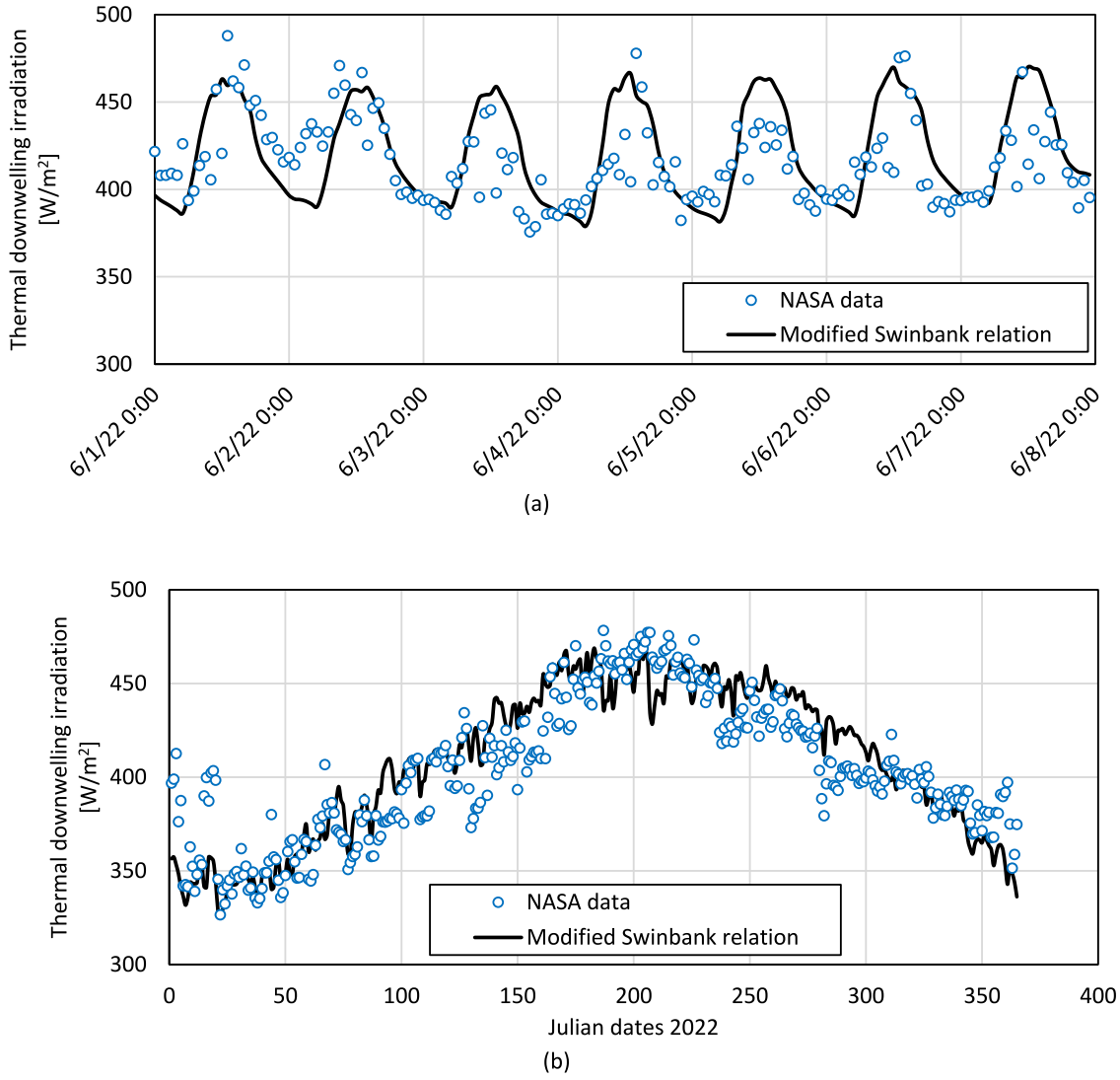


Fig. 5. Thermal downwelling irradiance value for (a) the first week of June 2022 and (b) the full year of 2022.

$$\dot{Q}_{r,PV} = \varepsilon_{PV,L}F_{PV,sky}\sigma A_{PV}\left(T_{PV}^4 - T_{sky}^4\right) + \varepsilon_{PV,L}F_{PV,grd}\sigma A_{PV}\left(T_{PV}^4 - T_{grd}^4\right) + \varepsilon_{PV,L}F_{PV,R}\sigma A_{PV}\left(T_{PV}^4 - T_R^4\right) + \varepsilon_{PV,L}F_{PV,b,grd}\sigma A_{PV}\left(T_{PV}^4 - T_{grd}^4\right) \quad (18)$$

Where h_t and h_b, are the air natural convection coefficient on the PV top-surface and bottom-side surface, respectively. A_{PV}, T_s, T_{amb}, σ and ε_{PV}, are the PV surface area, the PV surface temperature, the ambient temperature, the Stefan-Boltzmann constant, and the PV emissivity, respectively. $F_{PV,sky}$, $F_{PV,grd}$, $F_{PV,R}$ and $F_{PV,b,grd}$ are view factors and are defined with its value in Table 1.

The convection coefficient for top-surface (h_t) and bottom-surface (h_b) are calculated from Nusselt number relations from top- (Nu_t) and bottom- (Nu_b) surface using hot plate equations shown below.

$$h_t = Nu_t\frac{k_a}{L} \quad (19a)$$

$$h_b = Nu_b\frac{k_a}{L} \quad (19b)$$

$$Nu_t = 0.15\ Ra^{1/3} \quad (20a)$$

$$Nu_b = 0.1\ Ra^{1/3} \quad (20b)$$

$$Ra_L = Gr_L \bullet Pr = \frac{g\cos\theta\beta(T_{PV} - T_{\text{amb}})L^3}{\nu^2}Pr \quad (21)$$

Where Ra_L, Gr_L and Pr are Rayleigh, Grashof and Prandtl numbers, respectively. In addition, g, θ, β, k_a and ν represent gravity acceleration, the angle between the plate and normal axis, air thermal expansion, air thermal conductivity and air kinematic viscosity, respectively. Hence, the convection coefficients are calculated for the top- and bottom-surface of the PV.

At the radiator section, the heat is lost by convection ($\dot{Q}_{c,R}$) and radiation ($\dot{Q}_{r,R}$). By constructing the PV and the LHP from the same material, one can assume the same temperature for the PV panel and the LHP-evaporator. By adding radiative material paint to the LHP-condenser side, one can also assume the same temperature for the radiator and the LHP-condenser surface. The temperature difference between the evaporator (T_{PV}) and condenser (T_R) depends on the LHP design thermal resistance (R_{LHP}), which is mathematically expressed as follows:

$$T_R = T_{PV} - \dot{Q}_{LHP}/R_{LHP} \quad (22)$$

Hence, the convection and radiation heat transfer from the radiator (the two terms in equation (12)) are calculated, respectively, using the

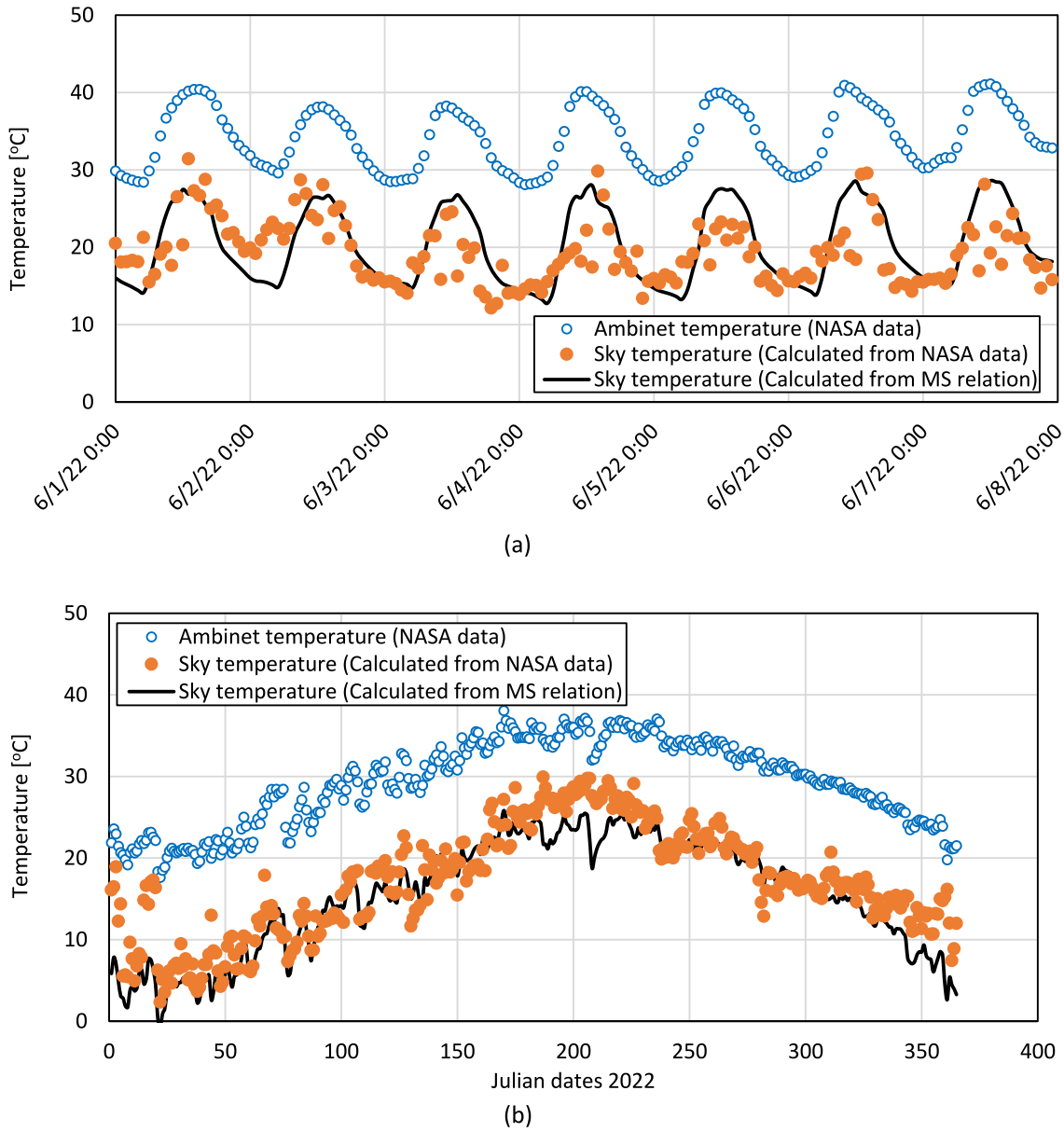


Fig. 6. Sky temperature calculated from modified Swinback relation and NASA data for (a) the first week of June 2022 and (b) the full year of 2022.

following equations:

$$\dot{Q}_{c,R} = h_t A_R (T_R - T_{amb}) + h_b A_R (T_R - T_{amb}) \tag{23}$$

$$\dot{Q}_{r,R} = \varepsilon_{R,L} F_{R,sky} \sigma A_R \left(T_R^4 - T_{sky}^4\right) + \varepsilon_{R,L} F_{R,grd} \sigma A_R \left(T_R^4 - T_{grd}^4\right) + \varepsilon_{R,L} F_{R,PV} \sigma A_R \left(T_R^4 - T_{PV}^4\right) + \varepsilon_{R,L} F_{R,b,grd} \sigma A_R \left(T_R^4 - T_{grd}^4\right) \tag{24}$$

Where h_t and h_b, are the air natural convection coefficient on the radiator top-surface and bottom-surface, respectively. A_R, T_R, T_{amb}, σ and $\varepsilon_{R,L}$, are the radiator surface area, the radiator surface temperature and the ambient temperature, the Stefan-Boltzmann constant, and the radiator longwave irradiance emissivity, respectively. $F_{R,sky}$, $F_{R,grd}$, $F_{R,PV}$ and $F_{R,b,grd}$ are view factors and are defined with values shown in Table 1.

2.5. Stand-alone PV panel

Fig. 2 shows a schematic of a single stand-alone PV panel that is cooled, mainly, by natural convection and radiation. The energy equation for this system is shown below:

$$\left[\alpha_{PV,S}(\dot{I}_{PV} + \dot{D}) + \alpha_{PV,L}\dot{q}_{sky}\right] = \dot{E}_{PV} + \dot{Q}_{c,PV} + \dot{Q}_{r,PV} \tag{25}$$

The electrical power produced ($\dot{E}_{PV}$), the heat lost by convection ($\dot{Q}_{c,PV}$) and the heat lost by radiation ($\dot{Q}_{r,PV}$) are calculated as presented in an earlier section.

2.6. Assumptions and numerical solution

The above set of equations are solved using engineering equation solver (EES) while implementing the following assumptions. The following input parameters: the solar irradiance reaching the ground ($\dot{G}$), ambient temperature, relative humidity, and thermal downwelling irradiation ($\dot{D}$) have been acquired from the NASA Langley Atmospheric Science Data Center (POWER Projects) [31]. Additional input data related to the PV panel, LHP thermal resistance and radiator emissivity are listed in Table 2 while the optical properties used in the simulation are listed in Table 3. In addition, the following assumptions have been

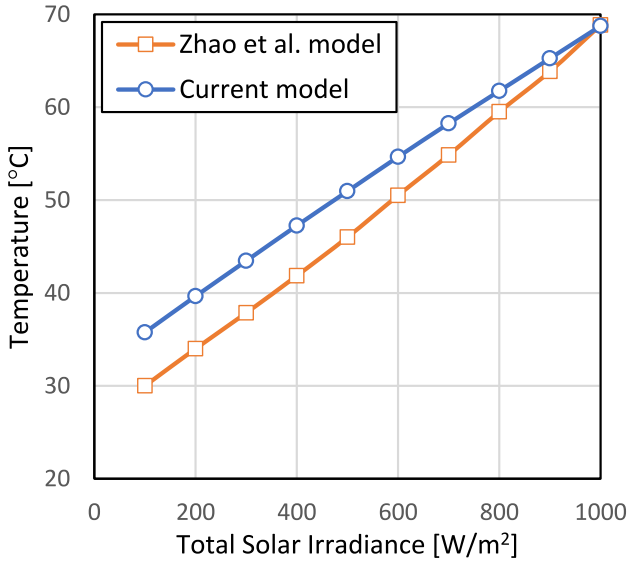


Fig. 7. Validating the model used in this study with literature by plotting PV panel temperature with different total solar irradiance.

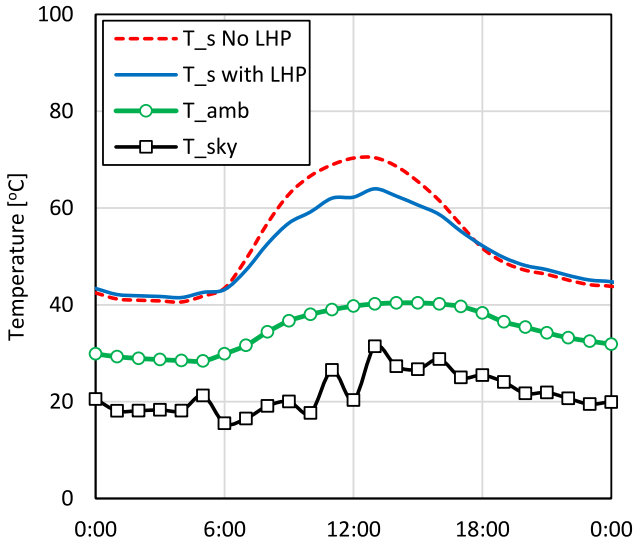


Fig. 8. The variation of temperature with time for PV panel with LHP-radiator cooling and without cooling on the first of June 2022.

used:

- The system is assumed to operate in quasi-steady state [15].
- The PV is assumed at uniform temperature T_{PV} and the radiator plate is assumed at uniform temperature T_R.
- The temperature difference between the top and bottom surface of PV is assumed negligible since cell thickness is submillimeter.
- The photovoltaic (PV) area (A_{PV}) is assumed to be equal to the radiator area (A_R). The PV area (A_{PV}) is set to 1 m^2 with width of 1 m and length of 1 m.
- It is assumed that the PV and the LHP-evaporator section are facing South, while the LHP-condenser is facing North.
- The contact resistance is assumed to be very small, hence PV surface temperature is assumed equal to the evaporator surface temperature.
- Simplified radiation heat transfer relations are adopted which are applicable for black body ($\varepsilon = 1$), while a more complex model is required when emissivity is not close to 1. Still above equations are considered acceptable since all the surroundings are assumed as black body meaning their emissivity is equal to 1.0, except for the PV surface which is set to emissivity of 0.89.
- The soil temperature is assumed to be equal to the surrounding ambient temperature. This is a rough assumption since the ground temperature lags behind the ambient temperature [43]. However, a reduced surface ground temperature will produce a lower PV temperature during daytime for both systems (single stand-alone PV and PV-LHP-RC system).
- The ratio of diffuse to total radiation is assumed to be 10 % since this value is not available in the POWER projects data and Sharjah is known for clear sky condition most of the year.
- The area of the return water pipes is assumed small and the pipe is presumed insulated.
- The diffuse radiation to the total radiation is assumed to be 10 %.
- The system is assumed to be modeled as two-dimensional since the PV panels are placed next to each other.
- The same tilt angle is used for the PV panel and the radiator surface.

3. Results and discussion

This section is divided into three parts. The first section discusses the results of solar irradiance, thermal downwelling and sky temperature values. The second section discusses the performance of the PV-LHP-RC system. The last section utilizes the developed model to conduct a parametric study and to estimate the impact of different parameters on the system.

3.1. Solar irradiance and sky temperature

To show the difference between “clear-sky surface” and “all-sky surface” irradiance, both irradiance values are plotted in Fig. 4 for the first week of June 2022. As shown in the figure, clear-sky surface irradiance shows a higher solar irradiance when compared to all-sky surface irradiance. The variation in these values arises from the influence of clouds, atmospheric particles, and other factors that can either scatter or absorb sunlight. Hence, in this study, only all-sky surface temperature has been used in the analysis.

The thermal downwelling irradiance represents the infrared radiation emitted by atmospheric components, such as greenhouse gases and clouds, which is directed downward and contributes to the overall heat transfer from the atmosphere to the Earth’s surface. The value of thermal downwelling has been calculated from the modified Swinbank relation and has been compared to the data produced from NASA Langley Atmospheric Science Data Center [31] as shown in Fig. 5. Fig. 5a and b shows the thermal downwelling irradiation in University City in Sharjah for the first week of June 2022, and for the whole year of 2022, respectively. Fig. 5a and b illustrate the effectiveness of the modified Swinbank relation as a powerful tool for predicting thermal downwelling irradiation in cases where such values are not readily available. Nevertheless, when accurate values of thermal irradiance downwelling are required, then one needs to obtain a measured data since as displayed in Fig. 5 there are discrepancies between the modified Swinbank relation and the experimental data. In this study, the values reported by POWER projects have been used in the analysis.

The sky temperature (T_{sky}) has been calculated from thermal downwelling irradiation $\left(\dot{q}_{sky}\right)$ and is plotted in Fig. 6. The thermal downwelling value has been calculated from the modified Swinbank relation and has been compared with the data collected from the NASA Langley Atmospheric Science Data Center (POWER Projects) [31]. Fig. 6a and b shows the sky temperatures and ambient temperatures for the first week of June 2022 and the daily averaged sky and ambient temperatures for the year 2022, respectively. As expected, the sky temperature is lower than the ambient temperature. Based on these

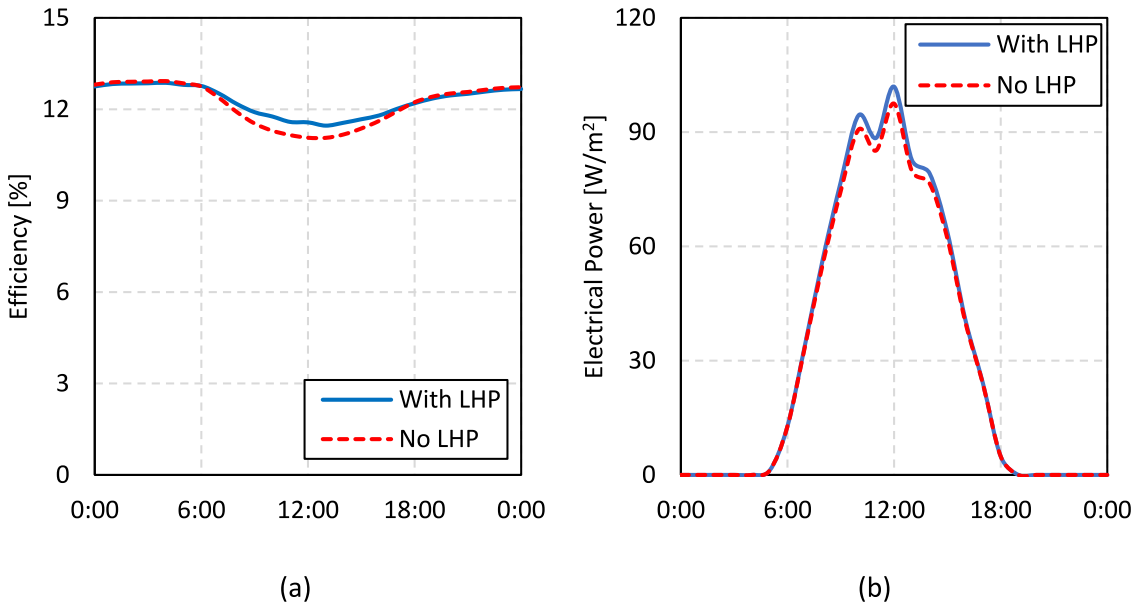


Fig. 9. PV electrical (a) efficiency and (b) power for 1 m^2 PV panel during the first of June 2022.

data, one can assume that the sky temperature is always lower than the ambient temperature by around 15 °*C*. Fig. 6a illustrates the fluctuation of sky temperature throughout the day, corresponding to the variations in ambient temperature resulting from sunrise and sunset. Fig. 6b demonstrates the annual variation in both daily-average ambient temperature and daily-average sky temperature, exhibiting fluctuations from higher values in summer to lower values in winter.

3.2. System performance evaluation

Validation of the results is an essential step for the acceptance of the proposed model. The novelty of the design made this step a challenging one. Nevertheless, a model has been developed for a single PV panel with no cooling system and has been compared to the work done by Zhao et al. [45] under similar conditions. The results of the current model are compared with the one developed by Zhao et al. [45] in Fig. 7. It is clear that the two models predict the same trend, but Zhao's model predicts a lower PV temperature. The reason for this discrepancy is due to the correlation used by Zhao et al. [45] to calculate the atmosphere emissivity, which depends on precipitable water vapor amount as the only input parameter [46], while this study uses the published data from NASA POWER projects. For the validation study, the modified Swinbank relation, which depends on atmospheric temperature, relative humidity, cloud coverage and cloud height, has been used. In the rest of the study, NASA POWER Projects have been used to evaluate the system for a more realistic model. As shown in Fig. 7, at a solar irradiance of 100 W/m^2, the current model results in a more realistic PV temperature which is higher than the reported ambient temperature of 30 °*C*, while the published work [45] predicted a PV temperature equal to the ambient temperature.

Fig. 8 shows the hourly ambient and sky temperatures for the first day in June 2022. The weather conditions of the first day of June 2022 have been used to calculate the PV panel surface temperature for two cases: PV without cooling system (T_s No LHP) and PV with LHP-radiator cooling system (T_s with LHP). As shown in the figure, the LHP-radiator cooling system provides a reduction in peak temperature by more than 10 °C. The figure also demonstrates that a reduced panel temperatures are attained over the daytime period while almost the same values are obtained during the nighttime period.

As mentioned earlier, PV panel temperature impacts its electrical efficiency. The impact of the PV panel temperature on the PV electric efficiency has been captured in Fig. 9a for the first day of June 2022. At peak conditions, a difference of 0.65 % in PV efficiency has been detected, which accounts for a 4.6 % increase in PV efficiency when compared to the PV reference efficiency. To quantify the benefit of the proposed LHP-Radiator cooling system, the PV electrical power per unit area has been plotted in Fig. 9b for both cases for the first day of June 2022. As depicted in the figure, the electricity output from a PV-LHP-RC system surpasses the standalone PV panel system.

To quantify the benefit of the proposed LHP-radiator cooling system, the amount of additional electrical energy (Wh) harvested per unit area and its percentage have been plotted per month as shown in Fig. 10a and b, respectively. The figure shows the energy lost due to the lack of a PV cooling system, or in other words, it shows the amount of additional electrical energy that one can gain by cooling the PV panel using the proposed LHP system. The amount of energy lost over the year is maximum during the summer season and lowest during the winter season which is expected. Fig. 10b shows that cooling can improve power harvesting by 7.9 % in May. that power saving (or PV power harvesting) can It is estimated that that the proposed LHP-radiator system can improve the annual electrical energy conversion by 3.4 % which translates to an additional annual energy of 8.7 kWh/m^2.

3.3. Parametric analysis

Different design parameters which can impact the PV-LHP-RC system include the radiator area, the convection coefficient value, the LHP thermal resistance and the emissivity of the radiator. The effect of these parameters has been examined at a specific time, which has been selected as 1 p.m. on the first of June 2022. The weather conditions at this time are shown in Table 4.

The size of the radiator has a strong impact on the PV panel temperature since it acts as means to dissipate the heat to surroundings. The effect of ratio of condenser-area to evaporator-area ($AR = A_R/A_{PV}$) has been examined at 1 p.m. on the first of June 2022 as shown in Fig. 11. As expected, as area ratio increases the PV panel temperature exponentially decreases. At area ratio of zero, the panel temperature approaches the simple PV panel condition without using LHP cooling system. As depicted in the figure, with area ratio of one, the temperature drops from 70.9 to 64.0 °*C*, while efficiency increases from 11.0 to 11.5 %.

Another important parameter is the convection coefficient of the PV backside which depends on the setup installation. If the system is

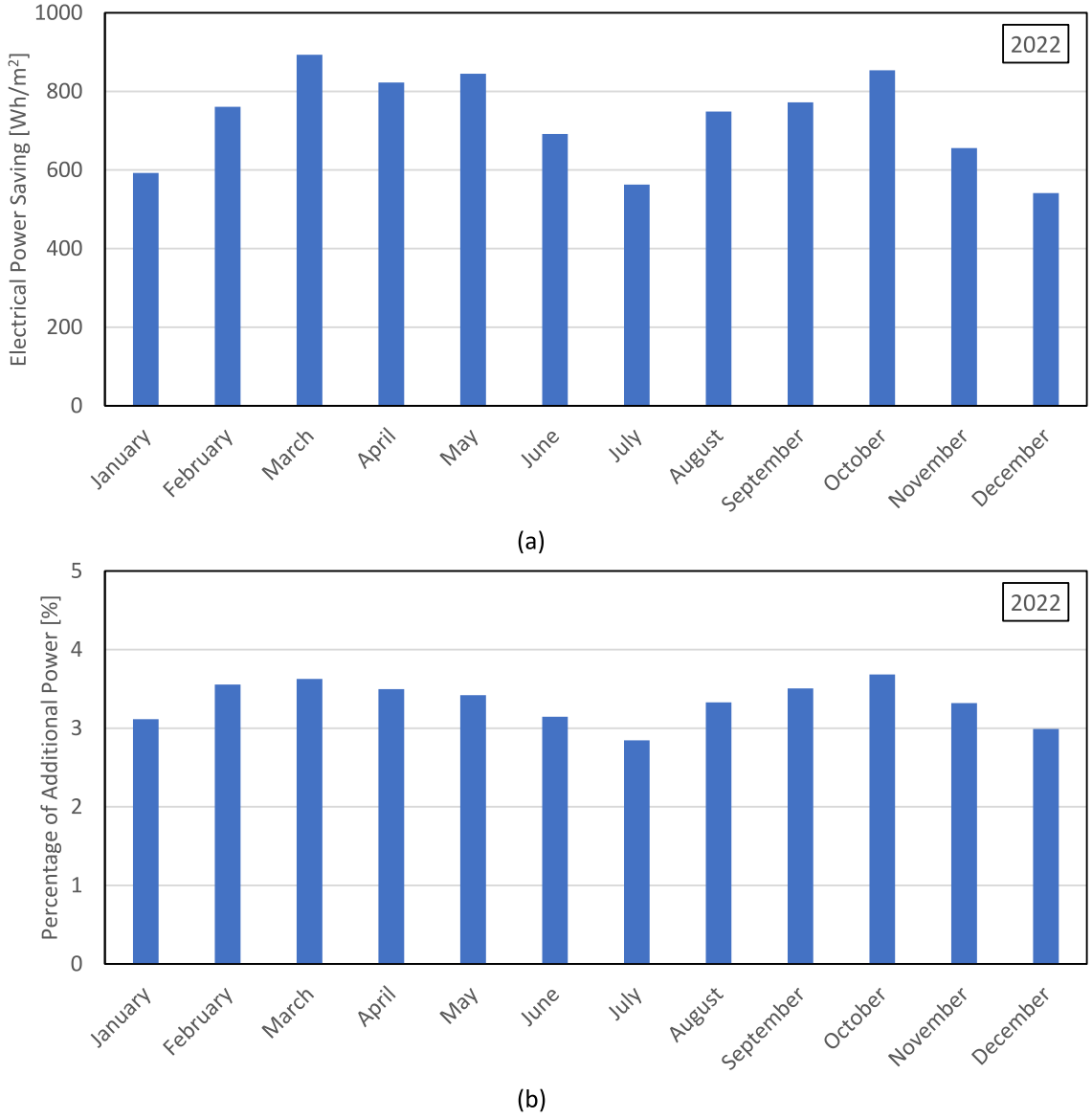


Fig. 10. An estimate of power saving per unit area when the LHP-RC system is used for (a) monthly saving and (b) percentage power saving for the year 2022.

Table 4
Weather conditions required in the calculation for the first of June 2022 at 1 p. m.

Date and time	Julian dates 2022	$\dot{q}_{sky}$	$\dot{G}$	T_{amb}	P_{amb}
		W/m²	W/m²	°C	kPa
6/1/22 13:00	152	488.0	770.4	40.2	99.4

installed on a roof where the backside convection and radiation are limited, then performance can drop dramatically. The impact of backside convection is shown in Fig. 12. It is clear that as convection heat transfer increases, the amount of heat removal increases which leads to lower PV surface temperature and higher electrical efficiency. Based on the first of June 2022 conditions, the convection coefficients on the front-side and backside were 13.54 and 3.16 $W/(m^2.^\circ C)$, which means the convection heat removal is dominated by frontside convection when compared to backside convection. Nevertheless, backside convection can decrease PV surface temperature by 11 $^\circ C$ when its value reaches 10 $W/(m^2.^\circ C)$. Under the weather conditions selected in Table 4, the backside convection is very small and it was able to shave only 0.5 $^\circ C$ from PV surface temperature.

The impact of the radiator shortwave and longwave irradiance emissivity/absorptivity are shown in Fig. 13a and b, respectively. Emissivity/absorptivity values can vary from 0 to 1. It is clear that shortwave emissivity ($\varepsilon_{R,S}$) has a stronger impact on the radiator performance compared to longwave emissivity ($\varepsilon_{R,L}$). Decreasing shortwave absorptivity near zero can decrease PV surface temperature by up to 4.5 $^\circ C$, while increasing longwave absorptivity near one can decrease PV surface temperature by less than 0.8 $^\circ C$.

The LHP thermal resistance plays an important role in moving heat from the PV panel to the radiator. As shown in Fig. 14 as LHP thermal resistance increases, the ability to move heat decreases, leading to an increase in PV surface temperature and consequently, a decrease in PV electrical efficiency.

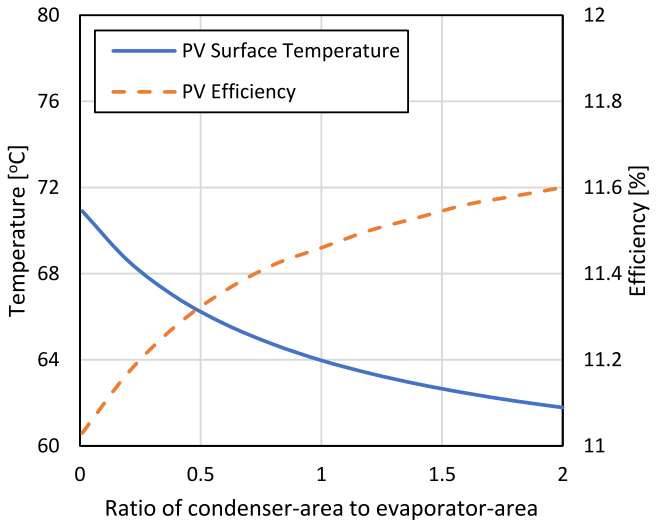


Fig. 11. The effect of ratio of condenser area to evaporator area on PV surface temperature and efficiency under conditions reported at 1 p.m. on the first of June 2022.

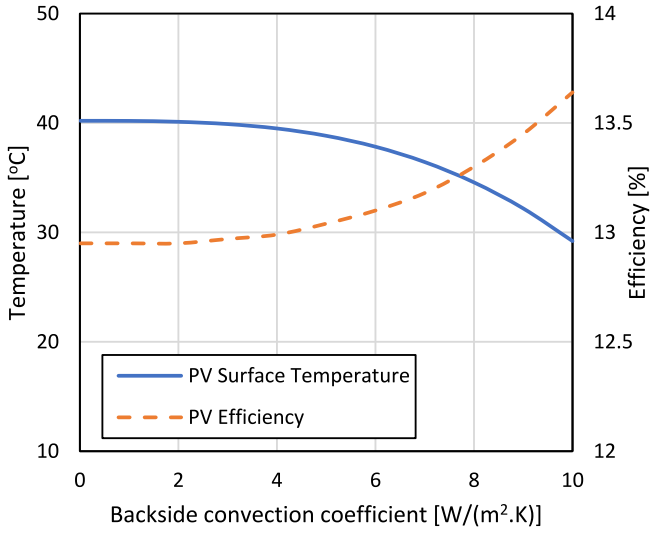


Fig. 12. The effect of the backside convection coefficient on PV surface temperature and efficiency under conditions reported at 1 p.m. on the first of June 2022.

4. Conclusions

A loop heat pipe (LHP) combined with a radiator system has been proposed as a cooling system for photovoltaic (PV) panels. The mathematical model developed for the PV-LHP-RC model has shown superior improvement when compared to stand-alone PV panel. The following key conclusions are drawn.

- The modified Swinbank relation provides an acceptable value for thermal downwelling irradiance. However, when modeling PV performance based on its temperature, reliance on experimental data is recommended.
- For Sharjah, it is appropriate to assume the sky temperature to be 15 °C below the ambient temperature.

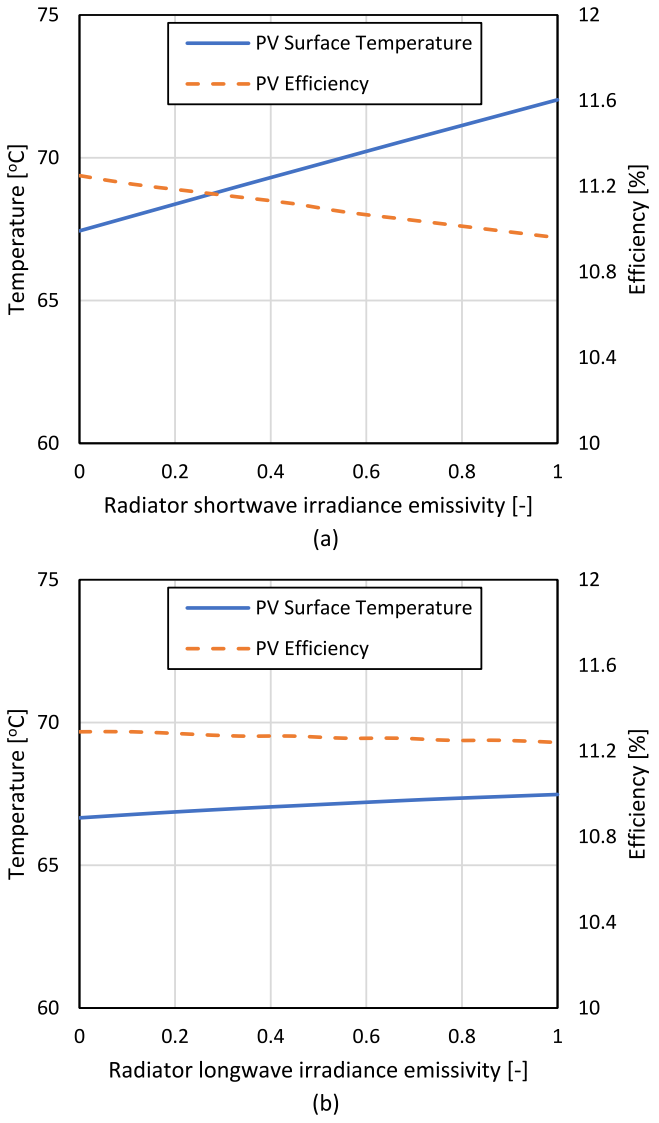


Fig. 13. The impact of the radiator (a) shortwave and (b) longwave irradiance emissivity/absorptivity on PV surface temperature and efficiency under conditions reported at 1 p.m. on the first of June 2022.

- PV cooling is needed since annual electrical energy lost could mount to 8.9 kWh/m^2 for selected PV panel.
- Stand-alone PV panel performance is notably deteriorated by elevated PV temperature during noontime and summer season.
- The proposed LHP-radiator cooling system offers a reduction in peak temperature by more than 10 °C, resulting in a 4.6 % improvement in PV efficiency compared to the reference PV efficiency.

CRediT authorship contribution statement

Mohammad O. Hamdan: Writing – original draft, Validation, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Frank M. Gerner:** Writing – review & editing, Methodology, Conceptualization. **Bassam A. Abu-Nabah:** Writing – review & editing, Supervision, Investigation, Funding acquisition. **Yahya Sheikh:** Writing – review & editing, Conceptualization. **Kareem Moursi:** Writing – review & editing, Visualization, Conceptualization.

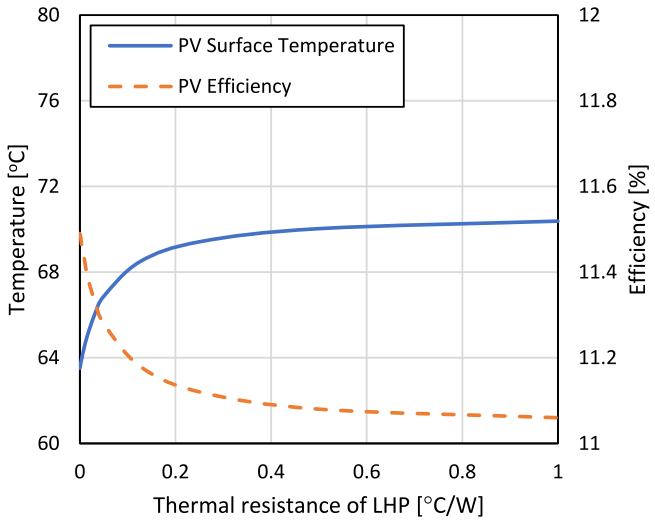


Fig. 14. The effect of LHP thermal resistance on PV surface temperature and efficiency under conditions reported at 1 p.m. on the first of June 2022.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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